Check for updates

INTERNATIONAL JOURNAL OF CLIMATOLOGY
Int. J. Climatol. **36**: 62–81 (2016)
Published online 8 April 2015 in Wiley Online Library
(wileyonlinelibrary.com) DOI: 10.1002/joc.4329

RMetS
Royal Meteorological Society

Comparison of homogeneity tests for temperature using a simulation study

Ceylan Yozgatligil* and Ceyda Yazici

Department of Statistics, Middle East Technical University, Ankara, Turkey

ABSTRACT: Homogeneity testing is one of the most important analyses in climate-related studies as it underpins the reliability of any inferences. The effects not directly related with climate are identified and removed from the meteorological variables, and then the obtained homogeneous variables are used to present an enhanced picture of the current situation and produce realistic forecasts based upon the variables. In this study, we investigate the performances of well-known homogeneity tests and introduce some tests that are not usually used for testing homogeneity via Monte Carlo simulation. We generate data using a normally distributed temperature variable. We consider both absolute and relative homogeneity tests (RHTest). Although relative tests are the best performing homogeneity tests, their results highly depend on the quality of the reference series. Consequently, they have to be used with at least one of the absolute tests in order to detect possible inhomogeneities in the reference series. Our results show that the relative tests which are standard normal homogeneity test, *F*-test for structural breaks with reference series, multiple change point detection method, namely and the RHTest have the best detection rates whereas the absolute tests, namely, Kruskal–Wallis, Pettitt and Friedman tests have the worst performances. The best performing absolute test is *F*-test for structural breaks.

KEY WORDS time series data; absolute homogeneity tests; relative homogeneity tests; unit root tests; stationarity tests; structural break tests

Received 11 March 2014; Revised 2 March 2015; Accepted 4 March 2015

1. Introduction

Given the impact of recent extreme weather events and the possibility that they may be attributable to climate change, serious attention is more warranted than ever towards analysing climate change. Of course, one of the ways to be prepared for an uncertain future is to ensure that we have the best tools possible to perform statistical analysis using meteorological variables. Testing the quality of the meteorological data is also critical as this bears greatly on the reliability of the results, and one of the most important quality control methods in climatological studies is testing the homogeneity of the meteorological data. In reality, finding completely homogeneous data is almost impossible, as the inevitable variability in the area surrounding a meteorological station usually affects the observed data. Inhomogeneity occurs when there are changes in the series that result from non-climatic causes. Some factors may hide the true climatic behaviour of the series, and consequently, the results obtained in the analysis of climate and hydrology related variables can be biased (Costa and Soares, 2009). Therefore, the temporal homogeneity of series is necessary in climatological research.

Inhomogeneity results from abrupt discontinuities, gradual or instant changes or changes in variability. Abrupt discontinuities, which are the most common type of inhomogeneities, are caused by changes in the location of the station, the instrumentation or the calculations of averages (Meteorological Organization World (WMO), 1996). Changes in the surroundings of the station, such as increasing urbanization, and changes in the instrumental attributes create gradual change in the series. Maintenance or calibration problems related to the apparatus can also occur. All of these have an impact on the characteristics of the series. Consequently, before starting any analysis, modelling or forecasting, these inhomogeneities must be detected and removed from the series, if possible.

Homogeneity tests can be divided into two groups: absolute tests that use the series itself and relative tests that use the information from neighbouring stations, known as reference stations. There are many absolute and relative tests available for testing homogeneity in the literature. However, in many of the previous studies on homogeneity, statistical tests developed for independent data have been used. Although in many homogeneity exercises, aggregated data are used to reduce or eliminate the autocorrelation problem, there might be still dependency within the series. Given that meteorological data such as total precipitation or average temperature are autocorrelated data, this study examined the performance of the homogeneity tests used in the literature on meteorological time series data and suggests some tests that take into account the autocorrelation structure of time series.

* Correspondence to: C. Yozgatligil, Department of Statistics, Middle East Technical University, Ankara 06800, Turkey. E-mail: ceylan@metu.edu.tr

© 2015 Royal Meteorological Society

Table 1. Percentage of inhomogeneous results when the mean shift of 0.5 unit increase is used at (1) the beginning of the series (break starts at year 5), (2) the middle of the series (break starts at year 27), and (3) the end of the series (break starts at year 53).

Test	Inhomogeneity detection (%) (1)		Breakpoint detection (%)	Inhomogeneity detection (%) (2)		Breakpoint detection (%)	Inhomogeneity detection (%) (3)		Breakpoint detection (%)
	Y_t	Y_t, shift		Y_t	Y_t, shift		Y_t	Y_t, shift	
SNHT	0.051	0.071	4→0.045 5→0.054	0.046	0.162	26→0.680 27→0.050	0.050	0.087	52→0.070 53→0.052
Pettitt	0.094	0.097	–	0.094	0.124	26→0.026 27→0.024	0.092	0.095	–
Buishand range	0.049	0.049	–	0.046	0.062	26→0.053 27→0.053	0.049	0.054	52→0.003 53→0.005
F-test	0.053	0.056	–	0.051	0.070	26→0.035 27→0.0	0.052	0.056	51→0.042
ADF	0.252	0.248	–	0.251	0.254	–	0.247	0.253	–
KPSS	0.050	0.051	–	0.047	0.073	–	0.050	0.050	–
Von Neumann	0.951	0.950	–	0.948	0.945	–	0.951	0.948	–
Kruskal–Wallis	0.042	0.042	–	0.040	0.049	–	0.043	0.047	–
Friedman	0.037	0.038	–	0.040	0.044	–	0.041	0.43	–
F-test with references	0.062	0.067	–	0.060	0.147	26→0.060 27→0.046	0.062	0.084	51→0.104
Bayes	–	–	4→0.0 5→0.0	–	–	26→0.0 27→0.0	–	–	–
Bayes with reference	–	–	4→0.0 5→0.0	–	–	26→0.0 27→0.0	–	–	–
TPR	0.062	0.062	4→0.003 5→0.004	0.0598	0.060	26→0.031 27→0.015	0.062	0.062	52→0.049 53→0.032
CauMe	0.310	0.316	4→0.042 5→0.325	0.308	0.327	26→0.013 27→0.013	0.314	0.327	52→0.201 53→0.160
RHTest	0.010	0.112	4→0.048 5→0.037	0.098	0.231	26→0.045 27→0.040	0.097	0.146	52→0.350 53→0.053
GAHMDI	0.241	0.280	4→0.080 5→0.081	0.251	0.385	26→0.050 27→0.037	0.241	0.292	52→0.040 53→0.045

In the literature, many methods are proposed to test the homogeneity of meteorological variables like precipitation and temperature (Ducre-Robitaille *et al.*, 2003; Staudt *et al.*, 2007). Potter (1981) uses the bivariate testing algorithm of Maronna and Yohai (1978) for detecting a mean shift using a reference series in precipitation series. Karl and Williams (1987) deal with the adjustment of the inhomogeneous series. They propose utilizing the Wilcoxon rank sum test and Student's *t*-test to identify inhomogeneities and mention the necessity of having accurate and complete historical information about the stations. In the adjustment, they make use of highly correlated reference stations and explain the importance of having a small variance in the adjustment procedure. Young (1993) suggests three different methods of interpolation to create reference series and uses a *t*-test to detect inhomogeneities. He also conducts a simulation study to check the performance of methods used in the study, and concludes that the major factors affecting the performance of the tests are the magnitude of the discontinuity and variance of the time series. Peterson and Easterling (1994) propose a method to obtain the reference series by using differences between successive years to decrease the effect of step-like discontinuities on the correlation coefficient. Tuomenvirta and Drebs (1994) suggest an additional requirement for adjusting a series, namely that each break identified by standard normal homogeneity test (SNHT) (Alexandersson and Moberg, 1997) should be supported by some physical evidence from the history of the station, which is sometimes impossible to accomplish because of incomplete metadata. Hanssen-Bauer and Førland (1994) and Tuomenvirta and Heino (1996) apply homogeneity analysis by using the SNHT method. Ozcelik (1996) uses the Kruskal–Wallis (KW) test, the Swed–Eisenhart runs test, and graphical analysis to test the homogeneity of precipitation series. Moberg and Alexandersson (1997) apply the SNHT through the homogenization of Swedish temperatures by using eight reference series with the highest spatial correlations for the first differenced series and employing the cutting algorithm introduced by Easterling and Peterson (1995). They also give recommendations on the optimal way of detecting multiple inhomogeneities using SNHT. Peterson *et al.* (1998) evaluate many methods to detect and adjust inhomogeneities in the meteorological series. Vincent (1998) proposes a method using four linear regression models to determine whether the series are homogeneous or not – if the residuals of the fitted models are white noise, the series is accepted as homogeneous.

© 2015 Royal Meteorological Society

10970088, 2016, 1, Downloaded from https://rmets.onlinelibrary.wiley.com/doi/10.1002/joc.4329 by University Of Texas At, Wiley Online Library on [16/10/2024]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Table 2. Percentage of inhomogeneous results when the mean shift of 1 unit increase is used at (1) the beginning of the series (break starts at year 5), (2) the middle of the series (break starts at year 27) and (3) the end of the series (break starts at year 53).

Test	Inhomogeneity detection (%) (1)		Breakpoint detection (%)	Inhomogeneity detection (%) (2)		Breakpoint detection (%)	Inhomogeneity detection (%) (3)		Breakpoint detection (%)
	Y_t	Y_t, shift		Y_t	Y_t, shift		Y_t	Y_t, shift	
SNHT	0.051	0.145	4→0.200 5→0.096	0.050	0.572	26→0.172 27→0.094	0.047	0.267	52→0.220 53→0.094
Pettitt	0.092	0.092	–	0.091	0.228	26→0.039 27→0.040	0.096	0.113	–
Buishand range	0.051	0.052	–	0.047	0.114	26→0.085 27→0.068	0.048	0.059	52→0.006 53→0.006
F-test	0.047	0.053	9→0.371	0.050	0.136	26→0.055 27→0.052	0.051	0.076	51→0.0
ADF	0.252	0.243	–	0.239	0.259		0.252	0.280	–
KPSS	0.046	0.047	–	0.045	0.147		0.045	0.064	–
Von Neumann	0.954	0.949	–	0.950	0.930		0.948	0.940	–
Kruskal–Wallis	0.043	0.043	–	0.042	0.072		0.041	0.055	–
Friedman	0.039	0.040	–	0.039	0.062		0.038	0.049	–
F-test with references	0.062	0.085	9→0.130	0.064	0.487	26→0.152 27→0.087	0.059	0.177	51→0.255
Bayes	–	–	4→0.0 5→0.0	–	–	26→0.0 27→0.0	–	–	52→0.0 53→0.0
Bayes with reference	–	–	4→0.0 5→0.0	–	–	26→0.0 27→0.0	–	–	52→0.0 53→0.0
TPR	0.062	0.062	4→0.022 5→0.177	0.064	0.064	26→0.080 27→0.048	0.064	0.064	52→0.086 53→0.067
CauMe	0.309	0.335	4→0.048 5→0.042	0.318	0.397	26→0.026 27→0.021	0.309	0.356	52→0.021 53→0.022
RHTest	0.092	0.169	4→0.107 5→0.566	0.102	0.604	26→0.104 27→0.096	0.091	0.298	52→0.113 53→0.091
GAHMDI	0.240	0.370	4→0.104 5→0.184	0.242	0.691	26→0.117 27→0.067	0.242	0.429	52→0.070 53→0.120

Tayanc *et al.* (1998) carry out a comparative evaluation by using Kruskal–Wallis and Wald–Wolfowitz methods to determine the inhomogeneous structure in a temperature series in Turkey.

When no appropriate reference stations exist, Sneyers *et al.* (1998) suggest using a nonparametric test operating only on the candidate series to search for break points, trends and other non-random behaviour from the series. Slonosky *et al.* (1999) use various methods to test the homogeneity of surface pressure series in Europe and state that the SNHT method shows satisfactory performance when a suitable reference series is obtained for comparison, evaluation and correction. Gonzales-Rouco *et al.* (2001) use SNHT after trimming the extreme values on 95 precipitation time series in southwestern Europe. Using SNHT, Tuomenvirta (2001, 2002) tests the homogeneity of temperature, precipitation and air pressure and applies adjustments on the inhomogenous series. Wijngaard *et al.* (2003) use the SNHT method, Buishand test, Pettitt test and Von Neumann test for testing the homogeneity of daily precipitation and temperature series. Begert *et al.* (2005) introduce THOMAS (tool for homogenization of monthly data series) with an application which senses the inhomogeneities and calculates the adjustments using 12 different homogeneity tests and least-squares fit for the adjustments. They find the inhomogeneities, most of which can be supported by the metadata, which is the crucial key for the homogeneity analysis. Mihajlovic (2006) uses the SNHT method to test the homogeneity of monthly total precipitation series used in the monitoring of a meteorological drought over the Pannonian part of Croatia. Karabork *et al.* (2007) apply the SNHT and Pettitt test to detect the inhomogeneity at precipitation gauging stations, and stations are considered inhomogeneous if at least one of the tests rejects the homogeneity hypothesis. Lund *et al.* (2007) propose a method based on Two-Phase Regression (TPR) to detect change points for periodic and autocorrelated time series. Reeves *et al.* (2007) compare and categorize eight methods to detect change points in climate series and propose a modification of (Lund and Reeves, 2002) (LR) and TPR with a common trend. Domonkos (2008b) provides a good comparison of homogenization methods. He includes 17 methods from the simple t-test to the multiple analysis of series for homogenization (MASH; Szentimrey, 1999). He finds that while the efficiency of the Mann-Kendall test and the t-test was low, the efficiencies of the other methods were around 70–76%. Syrakova and Stefanova (2009) apply the SNHT test to identify possible inhomogeneities on temperature series. They find that most of the inhomogeneities are

© 2015 Royal Meteorological Society

10970088, 2016, 1, Downloaded from https://rmets.onlinelibrary.wiley.com/doi/10.1002/joc.4329 by University Of Texas At, Wiley Online Library on [16/10/2024]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Table 3. Percentage of inhomogeneous results when the mean shift of 2 unit increase is used at (1) the beginning of the series (break starts at year 5), (2) the middle of the series (break starts at year 27) and (3) the end of the series (break starts at year 53).

Test	Inhomogeneity detection (%) (1)		Breakpoint detection (%)	Inhomogeneity detection (%) (2)		Breakpoint detection (%)	Inhomogeneity detection (%) (3)		Breakpoint detection (%)
	Y_t	Y_t, shift		Y_t	Y_t, shift		Y_t	Y_t, shift	
SNHT	0.048	0.529	4→0.473 5→0.115	0.050	0.998	26→0.452 27→0.124	0.048	0.854	52→0.541 53→0.126
Pettitt	0.093	0.110	–	0.097	0.578	26→0.116 27→0.065	0.089	0.177	52→0.001
Buishand range	0.049	0.060	–	0.050	0.352	26→0.176 27→0.101	0.050	0.094	52→0.036 53→0.011
F-test	0.053	0.074	4→0.0 5→0.0	0.053	0.430	26→0.002 27→0.002	0.051	0.159	51→0.282
ADF	0.246	0.225	–	0.256	0.325	–	0.255	0.364	–
KPSS	0.048	0.064	–	0.045	0.456	–	0.047	0.112	–
Von Neumann	0.950	0.930	–	0.950	0.850	–	0.950	0.911	–
Kruskal–Wallis	0.044	0.044	–	0.040	0.195	–	0.041	0.119	–
Friedman	0.038	0.038	–	0.038	0.165	–	0.041	0.087	–
F-test with references	0.063	0.190	9→0.290	0.060	0.991	26→0.440 27→0.117	0.063	0.665	51→0.530
Bayes	–	–	4→0.0 5→0.0	–	–	26→0.001 27→0.001	–	–	52→0.006 53→0.0
Bayes with reference	–	–	4→0.0 5→0.0	–	–	26→0.0 27→0.0	–	–	52→0.0 53→0.0
TPR	0.060	0.060	4→0.115 5→0.077	0.059	0.059	26→0.347 27→0.158	0.062	0.062	52→0.371 53→0.210
CauMe	0.308	0.406	4→0.117 5→0.061	0.306	0.660	26→0.070 27→0.044	0.306	0.488	52→0.068 53→0.050
RHTest	0.094	0.438	4→0.191 5→0.164	0.108	0.980	26→0.220 27→0.220	0.099	0.776	52→0.214 53→0.198
GAHMDI	0.241	0.650	4→0.142 5→0.394	0.254	0.976	26→0.334 27→0.110	0.232	0.803	52→0.324 53→0.110

caused by station relocations. Firat *et al.* (2010) use SNHT, (Swed–Eisenhart) runs test and Pettitt test to detect inhomogeneities in precipitation series and find that about 22% of the series are inhomogeneous but they could not support these findings with metadata. They also state that SNHT and Pettitt tests are more sensitive in determining inhomogeneity at gauging stations. Sahin and Cigizoglu (2010) use the SNHT, the Buishand range test, the Pettitt test and the Von Neumann ratio test on temperature, total precipitation and relative humidity series. Domonkos (2011) compares the performances of multiple linear regression (MLR; Vincent, 1998) and the second version of SNHT (Alexandersson and Moberg, 1997) with detection methods applying the always step function model, namely with MASH and PRODIGE (Caussinus and Mestre, 2004). His results show that Caussinus–Mestre (CauME) method and MASH are the most powerful techniques in identifying and correcting inhomogeneties in meteorological series. Sugahara *et al.* (2012) evaluate the effect of the replacement of instruments and of urbanization on meteorological data. They use the *F*-test for structural breaks (multiple breaks linear regression model), SNHT and Bayesian technique. To test the effect of urbanization, they use Kendall's tau test. Li and Lund (2012) estimate the number of change points and their locations by using genetic algorithms (GA). They use a minimum description length (MDL) criterion and then introduce GA to detect the change points. Toreti *et al.* (2012) propose a multiple change point detection method, namely GAHMDI, which is based on a GA and hidden Markov models. Their simulation study shows that GAHMDI performs better than Caussinus and Mestre (2004), the SNHT (Alexandersson and Moberg, 1997) and the RHTest (Wang *et al.*, 2007; Wang 2008a, 2008b). It also accounts for the variance changes but not the autocorrelation in the series. Domonkos (2013) examines the methodology of quantifying the efficiency of homogenization methods using simulated series. To measure efficiency, he uses root mean-squared error (RMSE), centered RMSE, hit rate, false alarm rate and their combinations. This article focuses on homogenization and not the detection of inhomogeneities but discusses the usefulness of the detection methods in the homogenization procedure.

The performance comparison of the homogeneity tests on the meteorological data are examined by several studies. In the study of Ducre-Robitaille *et al.* (2003), eight different methods are used, and shown that the SNHT and multiple regression methods perform the best. However, they generate data for annual mean temperature series from an autoregressive model with order 1 without

© 2015 Royal Meteorological Society

10970088, 2016, 1, Downloaded from https://rmets.onlinelibrary.wiley.com/doi/10.1002/joc.4329 by University Of Texas At, Wiley Online Library on [16/10/2024]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Table 4. Percentage of inhomogeneous results when the sudden decrease is used at (1) the beginning of the series,(2) the middle of the series (3) the end of the series.

Test	Inhomogeneity detection (%) (1)		Breakpoint detection (%)	Inhomogeneity detection (%) (2)		Breakpoint detection (%)	Inhomogeneity detection (%) (3)		Breakpoint detection (%)
	Y_t	Y_t, shift		Y_t	Y_t, shift		Y_t	Y_t, shift	
SNHT	0.049	0.050	4→0.027 5→0.021	0.050	0.049	26→0.008 27→0.002	0.047	0.048	52→0.030 53→0.022 3→0.022
Pettitt	0.096	0.095	–	0.094	0.094	26→0.014 27→0.011	0.095	0.093	–
Buishand range	0.051	0.050	–	0.047	0.047	26→0.046 27→0.040	0.053	0.053	52→0.001 53→0.002
F-test	0.053	0.053	–	0.051	0.051	26→0.029 27→0.017	0.051	0.052	51→0.049
ADF	0.250	0.250	–	0.246	0.247	–	0.255	0.255	–
KPSS	0.047	0.046	–	0.045	0.045	–	0.048	0.048	–
Von Neumann	0.947	0.947	–	0.949	0.948	–	0.950	0.949	–
Kruskal–Wallis	0.041	0.040	–	0.041	0.040	–	0.041	0.040	–
Friedman	0.036	0.036	–	0.039	0.039	–	0.039	0.038	–
F-test with references	0.061	0.060	9→0.302	0.066	0.063	26→0.018 27→0.017	0.063	0.067	51→0.060
Bayes	–	–	4→0.001 5→0.0	–	–	26→0.0 27→0.0	–	–	52→0.0 53→0.0
Bayes with reference	–	–	4→0.001 5→0.0	–	–	26→0.0 27→0.0	–	–	52→0.0 53→0.0
TPR	0.062	0.062	4→0.019 5→0.019	0.060	0.060	26→0.018 27→0.018	0.061	0.061	52→0.017 53→0.030
CauMe	0.313	0.331	4→0.045 5→0.039	0.305	0.308	26→0.016 27→0.008	0.318	0.309	52→0.014 53→0.011
RHTest	0.086	0.093	4→0.036 5→0.036	0.101	0.098	26→0.018 27→0.018	0.094	0.095	52→0.021 53→0.010
GAHMDI	0.241	0.245	4→0.055 5→0.056	0.236	1.000	26→0.845 27→0.000	0.236	1.000	52→0.738 53→0.0

any justification from actual temperature series. Moreover, they only consider two scenarios of inhomogeneity: fixed and random mean shifts. Domonkos (2008a) also compares 15 detection and correction methods. He concludes that Easterling and Peterson (1995) method is the best in detection skill, while nonparametric tests produce worse results. Another study on the performances of the tests is the study by Costa and Soares (2009). They use the SNHT, the Buishand range test, the Pettit test and the geostatistical technique introduced by the authors. They show that the geostatistical technique is a good method of detecting inhomogeneities. Venema *et al.* (2012) make a comprehensive comparison of the detection and homogenization algorithms with applications on real and simulated data. Moreover, Guijarro (2013) compares six methods on running windows of the series generated from normally distributed terms and conclude that *t*-test and SNHT are the best tests. Yozgatligil *et al.* (2010, 2011) and Yazici *et al.* (2012) perform a simulation study on the validity of homogeneity tests. Their argument is the unsuitability of the homogeneity tests that are not appropriate for time series data. Yozgatligil *et al.* (2011) use KW, Friedman test and Kwiatkowski, Phillips, Schmidt and Shin stationarity test (KPSS) and conclude that the KPSS test gives better results in detecting inhomogeneities. Yazici *et al.* (2012) consider SNHT, KW, Buishand Range, Pettitt, Von Neumann and F-test for structural breaks (or Chow test) and, as a result of their simulation study, and suggest that the SNHT and F-test for Structural Breaks are the best two tests and that Von Neumann is the worst-performing test. All the previous studies generate data for temperature series. In this study, we extend the study of Yazici *et al.* (2012) and apply the most popular homogeneity tests on the temperature series in the simulation study. We have selected several well-known homogeneity tests:

- SNHT (Alexandersson, 1986),
- the Buishand range test (Buishand, 1982),
- the Pettitt test (Pettitt, 1979),
- the Von Neumann ratio test (Von Neumann, 1941),
- the KW test (Kruskal and Wallis, 1952),
- the TPR
- the RHTest (Wang, 2008a, 2008b)
- CauMe (Caussinus and Mestre, 2004)
- the GAHMDI (Toreti *et al.*, 2012)

Also, the tests that are specifically used for time series data in statistics literature:

© 2015 Royal Meteorological Society

10970088, 2016, 1, Downloaded from https://rmets.onlinelibrary.wiley.com/doi/10.1002/joc.4329 by University Of Texas At, Wiley Online Library on [16/10/2024]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Table 5. Percentage of inhomogeneous results when the gradual increase is used at (1) the beginning of the series (between 5th and 10th realizations), (2) the middle of the series (between 27th and 32nd realizations) (3) the end of the series (between 53rd and 58th realizations).

Test	Inhomogeneity detection (%) (1)		Breakpoint detection (%)	Inhomogeneity detection (%) (2)		Breakpoint detection (%)	Inhomogeneity detection (%) (3)		Breakpoint detection (%)
	Y_t	Y_t, shift		Y_t	Y_t, shift		Y_t	Y_t, shift	
SNHT	0.048	0.053	4→0.022 5→0.020	0.051	0.052	26→0.022 27→0.006	0.053	0.053	51→0.032 52→0.022
Pettitt	0.087	0.087	–	0.094	0.094	26→0.017 27→0.014	0.093	0.092	–
Buishand range	0.045	0.046	4→0.0 5→0.002	0.048	0.048	26→0.028 27→0.016	0.048	0.048	52→0.006 53→0.004
F-test	0.050	0.052	–	0.055	0.053	26→0.020 27→0.022	0.052	0.052	51→0.046
ADF	0.249	0.249	–	0.247	0.250	–	0.255	0.255	–
KPSS	0.044	0.045	–	0.048	0.048	–	0.048	0.048	–
Von Neumann	0.949	0.949	–	0.949	0.948	–	0.949	0.949	–
Kruskal–Wallis	0.038	0.040	–	0.042	0.043	–	0.043	0.043	–
Friedman	0.038	0.040	–	0.040	0.040	–	0.038	0.038	–
F-test with references	0.060	0.063	4→0.0 5→0.0	0.063	0.064	26→0.017 27→0.015	0.062	0.063	51→0.040
Bayes	–	–	4→0.0 5→0.0	–	–	26→0.0 27→0.0	–	–	52→0.0 53→0.0
Bayes with reference	–	–	4→0.0 5→0.0	–	–	26→0.0 27→0.0	–	–	52→0.0 53→0.0
TPR	0.061	0.061	4→0.006 5→0.011	0.064	0.064	26→0.017 27→0.234	0.064	0.064	51→0.023 52→0.017
CauMe	0.309	0.310	4→0.030 5→0.028	0.318	0.317	26→0.0081 27→0.01	0.318	0.317	52→→0.013 53→0.013
RHTest	0.104	0.113	4→0.047 5→0.021	0.093	0.096	26→0.022 27→0.026	0.090	0.098	52→0.018 53→0.008
GAHMDI	0.229	0.257	4→0.061 5→0.056	0.230	0.249	26→0.002 27→0.003	0.237	0.237	52→0.0 53→0.0

- the F-test for structural breaks (Chow, 1960)
- the KPSS stationarity test (Kwiatkowski *et al.*, 1992)
- the Augmented Dickey-Fuller (ADF) unit root test (Said and Dickey, 1984)
- the Bayesian test (Barry and Hartigan, 1992, 1993)
- the Friedman test (Friedman, 1937), which considers data dependency partially

A Bayesian test to find inhomogeneities can be seen in the studies of Ducre-Robitaille *et al.* (2003) and Sugahara *et al.* (2012). The F-test for structural breaks (Sugahara *et al.*, 2012), ADF unit root test and KPSS stationarity test are not commonly used in homogeneity detection analysis.

This article is organized as follows: Section 1 details the purpose of the study along with a review of the literature that covers the studies about the homogeneity analysis of meteorological variables and about the comparison of the performance of homogeneity tests. Section 2 describes the selected absolute and relative tests. Section 3 discusses the simulation setup, applications, and the performances of all the tests under homogeneous reference series and relative tests with an inhomogeneous reference series. In Section 4, the findings of the study are presented, followed by the concluding remarks and brief statement on further studies.

2. Description of homogeneity tests

In this section, the relative and absolute tests that are used in this study are described. The most frequently used homogeneity tests in the literature were selected, as well as a few other less well-known tests for homogenization purposes.

2.1. Absolute tests

2.1.1. Kruskal–Wallis test

The KW test (Kruskal, 1952; Kruskal and Wallis, 1952) is a well-known nonparametric test used to compare two or more independent groups of sampled data and is one of the most applied for checking the homogeneity. However one of the assumptions of the KW test is that the observations are drawn randomly and independently from their respective populations. It is less suitable for testing the mean differences within time series.

In the computations, initially all the series are combined into a single dataset and then the rank score of each small

© 2015 Royal Meteorological Society

10970088, 2016, 1, Downloaded from https://rmets.onlinelibrary.wiley.com/doi/10.1002/joc.4329 by University Of Texas At, Wiley Online Library on [16/10/2024]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Table 6. Percentage of inhomogeneous results when the increasing trend with slope (1) 0.00075 (monthly) and (2) 0.0033 (monthly) are applied.

Test	Inhomogeneity detection (%) (1)		Inhomogeneity detection (%) (2)	
	Y_t	Y_t, shift	Y_t	Y_t, shift
SNHT	0.048	0.116	0.047	0.915
Pettitt	0.092	0.113	0.093	0.378
Buishand range	0.046	0.052	0.047	0.161
F-test	0.052	0.065	0.051	0.277
ADF	0.252	0.252	0.254	0.254
KPSS	0.045	0.061	0.045	0.312
Von Neumann	0.947	0.946	0.945	0.900
Kruskal–Wallis	0.040	0.044	0.042	0.107
Friedman	0.037	0.039	0.039	0.090
F-test with references	0.061	0.113	0.062	0.844
Bayes	–	–	–	–
Bayes with reference	–	–	–	–
TPR	0.062	0.062	0.060	0.060
CauMe	0.309	0.325	0.308	0.524
RHTest	0.095	1.000	0.098	1.000
GAHMDI	0.238	0.298	0.240	0.234

series is found separately from the rank order of the pooled data. Finally, the test statistic H is found by

$$H = \frac{12}{n(n+1)} \sum_{i=1}^{k} \frac{R_i^2}{n_i} - 3(n+1)$$

where n is the total sample size in the pooled data, n_i is the sample size, R_i is the rank of the ith series and k is the number of groups. In the climate applications of this test, a series is divided into several groups covering 7 or 10 years in each one (Turkes, 1999; Turkes *et al.*, 2008). Therefore, there is dependence within and between groups. The KW test gives little information about the probable date for a shift in the median and no information about the magnitude of the break.

2.1.2. *Friedman test*

The KW test does not take into account the autocorrelation in the series so we wanted to look at the performance of the Friedman Test which is the nonparametric version of the repeated Analysis of Variance test (Friedman, 1937). Then, we computed the sum of ranks within each column by the following expression F.

$$F = \frac{12}{bc(c+1)} \sum_{i=1}^{c} R_i^2 - 3b(c+1)$$

where R_i denotes the underlying rank for the ith series ($i = 1, \ldots, t$), b and c indicate the number of columns and rows of the data matrix, respectively. For the climatological dataset, b can represent the number of blocks that data separated in and c can be denoted as the number of time points. In this test, it is assumed that blocks are dependent. Finally, the significance of F is determined by comparing it with the test specific critical value (Sprent and Smeeton, 2007; Corder and Foreman, 2009).

2.1.3. *Buishand range test*

This is a parametric test which assumes an independent and identically normally distributed series (Wijngaard *et al.*, 2003). This test is more sensitive to breaks in the middle of a time series (Hawkins, 1977; Wijngaard *et al.*, 2003) and uses the adjusted partial sums which are defined as

$$S_0^* = 0 \text{ and } S_k^* = \sum_{i=1}^{k} \left(Y_i - \overline{Y}\right), \quad k = 1, 2, \ldots, n$$

where Y_i is the ith observation. If the series is homogeneous, then S_k^* fluctuates around zero. If there is a break in year K, then S_k^* reaches a maximum (negative shift) or minimum (positive shift) near the year $k = K$. The test statistic is defined as

$$\text{Range} = \left[\max S_k^* - \min S_k^*\right] / s$$

where s is the sample standard deviation. The critical values for $\text{Range}/\sqrt{n}$ can be found in the study by Buishand (1982).

2.1.4. *Pettitt test*

The Pettitt test is a nonparametric test. The null hypothesis states that a series is independent and has identically distributed random quantities, and the alternative hypothesis is the existence of stepwise shift in the mean. The ranks $(R_1, R_2, \ldots, R_n)$ of the original series, $Y_1, Y_2, \ldots, Y_n$ are used to calculate the test statistic. The ranking approach of this test implies that it is less sensitive to outliers than the other tests (Wijngaard *et al.*, 2003). This test is also sensitive to breaks in the middle of a time series (Hawkins, 1977; Wijngaard *et al.*, 2003). The test statistic is calculated as

$$Z_k = 2\sum_{i=1}^{k} R_i - k(n+1), \quad k = 1, \ldots, n$$

If a break occurs in year K, then the statistic is maximal or minimal near the year $k = K$. So, if the break occurs, then $Z_K = \max Z_k$. The critical values of the test are given in the study by Pettitt (1979).

2.1.5. *Von Neumann ratio test*

This test is a parametric test which does not give information on the year of the break. It uses the ratio of the mean square successive (year to year) difference to the variance. The Von Neumann ratio test is complementary to other tests because of its sensitivity to departures from homogeneity due to its nature (Wijngaard *et al.*, 2003). The Von Neumann ratio is defined as the ratio of the mean square successive difference to the variance

$$N = \sum_{i=1}^{n-1} \left(Y_i - Y_{i+1}\right)^2 / \sum_{i=1}^{n} \left(Y_i - \overline{Y}\right)^2$$

When the sample is homogeneous, the expected value is $N = 2$. Small values of N indicate inhomogeneity.

© 2015 Royal Meteorological Society

10970088, 2016, 1, Downloaded from https://rmets.onlinelibrary.wiley.com/doi/10.1002/joc.4329 by University Of Texas At, Wiley Online Library on [16/10/2024]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Table 7. Percentage of inhomogeneous results when the mean shift of 0.5 unit increase is used at (1) the beginning of the series (break starts at year 5), (2) the middle of the series (break starts at year 27), and (3) the end of the series (break starts at year 53) when there is a trend in the series.

Test	Inhomogeneity detection (%) (1)		Breakpoint detection (%)	Inhomogeneity detection (%) (2)		Breakpoint detection (%)	Inhomogeneity detection (%) (3)		Breakpoint detection (%)
	Y_t	Y_t, shift		Y_t	Y_t, shift		Y_t	Y_t, shift	
SNHT	0.051	0.204	4→0.050 5→0.046	0.048	0.456	26→0.075 27→0.058	0.047	0.270	51→0.051 52→0.066
Pettitt	0.096	0.124	–	0.088	0.196	26→0.027 27→0.030	0.092	0.133	–
Buishand range	0.049	0.054	4→0.001 5→0.005	0.045	0.088	26→0.060 27→0.060	0.047	0.061	52→0.011
F-test	0.052	0.073	–	0.049	0.119	26→0.038 27→0.029	0.049	0.083	51→0.074
ADF	0.251	0.249	–	0.257	0.262	–	0.248	0.258	–
KPSS	0.045	0.071	–	0.043	0.127	–	0.046	0.083	–
Von Neumann	0.950	0.945	–	0.947	0.933	–	0.953	0.944	–
Kruskal–Wallis	0.041	0.045	–	0.038	0.061	–	0.038	0.049	–
Friedman	0.038	0.041	–	0.038	0.505	–	0.036	0.045	–
F-test with references	0.062	0.159	–	0.058	0.371	26→0.002 27→0.002	0.062	0.206	51→0.095
Bayes	–	–	4→0.0 5→0.0	–	–	26→0.0 27→0.0	–	–	52→0.0 53→0.0
Bayes with reference	–	–	4→0.0 5→0.0	–	–	26→0.0 32→0.0	–	–	52→0.0 53→0.0
TPR	0.060	0.060	4→0.013 5→0.006	0.058	0.058	26→0.029 27→0.032	0.067	0.067	52→0.041 53→0.041
CauMe	0.310	0.340	4→0.036 5→0.028	0.310	0.390	26→0.012 27→0.014	0.310	0.636	52→0.133 53→0.010
RHTest	0.093	1.000	4→0.038 5→0.047	0.099	1.000	26→0.044 27→0.038	0.089	1.000	52→0.0 53→0.0
GAHMDI	0.242	0.302	4→0.053 5→0.066	0.230	0.303	26→0.037 27→0.020	0.237	0.302	52→0.0 53→0.0

2.1.6. *KPSS test*

Most of the homogeneity tests do not take the autocorrelation structure of the meteorological data into account when testing the homogeneity of the series (Yazici *et al.*, 2011). Stationarity tests, which are developed specifically for time series data, can also be used to identify the existence of inhomogeneity within the series. As the test is not a location specific test, if the series is not stationary, another test can be applied to find the exact change point. The KPSS test (Kwiatkowski *et al.*, 1992) is commonly used to test the existence of trend within the series. In the case of mean shift, the test may detect the positive or negative shift as a trend in the series. The KPSS test is developed by regressing Y_t on a constant and constructing the ordinary least square residuals $\varepsilon = (\varepsilon_1, \varepsilon_2, \cdots, \varepsilon_n)'$. Then, the partial sum of the residuals is obtained as $e_t = \sum_{i=1}^{t} \varepsilon_i$. The KPSS test statistic is given by

$$\mathrm{KPSS} = n^{-2} \sum_{t=1}^{n} \frac{e_t}{\widehat{\sigma}^2}$$

where $\widehat{\sigma}^2$ is the estimate of the long-run variance of the residuals. If the series is not stationary, i.e. inhomogeneous, then the null hypothesis should be rejected. An asymptotic distribution of the test statistic uses the standard Brownian bridge.

2.1.7. *ADF test*

This test is used for testing the stationarity of the time series. Similar to KPSS test, ADF searches for the trend within the series. We added this test to the list of homogeneity tests because it may detect if the mean of the series is changing over time. Considering the pth order autoregressive process, AR(p),

$$Y_t = \theta_0 + \phi_1 Y_{t-1} + \cdots + \phi_p Y_{t-p} + a_t$$

where θ_0 is a function of process mean, ϕ_is are model parameters and a_t is a White Noise process with zero mean and constant variance. The augmented Dickey-Fuller (DF) test (Said and Dickey, 1984) is the generalized form of the DF unit root test. To be able to perform the test, the process should be written in backshift operator form, $(1 - \phi_1 B - \cdots - \phi_p B^p) Y_t = \theta_0 + a_t$. If some of roots of the polynomial $(1 - \phi_1 B - \cdots - \phi_p B^p)$ are 1, then the series is not stationary. To test this, the following ADF test equation is used

$$\Delta Y_t = \delta Y_{t-1} + \sum_{j=1}^{p-1} \phi_j \Delta Y_{t-j} + \theta_0 + a_t.$$

when $\delta = 1$, we have a unit root, i.e. the series is not stationary. The t-test statistic is used but the statistic has a non-standard distribution. The critical values can be obtained from the study by Said and Dickey (1984).

© 2015 Royal Meteorological Society

10970088, 2016, 1, Downloaded from https://rmets.onlinelibrary.wiley.com/doi/10.1002/joc.4329 by University Of Texas At, Wiley Online Library on [16/10/2024]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Table 8. Percentage of inhomogeneous results when the mean shift of 1 unit increase is used at (1) the beginning of the series (break starts at year 5); (2) the middle of the series (break starts at year 27), and (3) the end of the series (break starts at year 53) when there is a trend in the series.

TEST	Inhomogeneity detection (%) (1)		Breakpoint detection (%)	Inhomogeneity detection (%) (2)		Breakpoint detection (%)	Inhomogeneity detection (%) (3)		Breakpoint detection (%)
	Y_t	Y_t, shift		Y_t	Y_t, shift		Y_t	Y_t, shift	
SNHT	0.049	0.333	4→0.014 5→0.070	0.051	0.860	26→0.186 27→0.090	0.050	0.537	52→0.164 53→0.082
Pettitt	0.089	0.132	–	0.090	0.336	26→0.056 27→0.041	0.092	0.170	–
Buishand range	0.047	0.059	–	0.048	0.163	26→0.103 27→0.069	0.045	0.071	52→0.016
F-test	0.049	0.083	–	0.045	0.220	26→0.074 27→0.052	0.049	0.118	51→0.126
ADF	0.025	0.245	–	0.250	0.270	–	0.247	0.276	–
KPSS	0.044	0.080	–	0.043	0.246	–	0.048	0.111	–
Von Neumann	0.952	0.943	–	0.950	0.906	–	0.952	0.933	–
Kruskal–Wallis	0.042	0.045	–	0.043	0.100	–	0.041	0.073	–
Friedman	0.036	0.039	–	0.039	0.086	–	0.037	0.059	–
F-test with references	0.062	0.207	–	0.059	0.778	26→0.167 27→0.090	0.065	0.397	51→0.211
Bayes	–	–	4→0.0 5→0.0	–	–	26→0.0 27→0.0	–	–	52→0.0 53→0.0
Bayes with reference	–	–	4→0.0 5→0.0	–	–	26→0.0 27→0.0	–	–	52→0.0 53→0.0
TPR	0.061	0.061	4→0.030 5→0.016	0.058	0.058	26→0.010 27→0.061	0.063	0.063	52→0.081 53→0.067
CauMe	0.313	0.364	4→0.050 5→0.040	0.309	0.485	26→0.030 27→0.024	0.300	0.406	52→0.029 53→0.029
RHTest	0.111	1.000	4→0.083 5→0.089	0.095	1.000	26→0.213 27→0.033	0.105	1.000	52→0.0 53→0.0
GAHMDI	0.240	0.298	4→0.078 5→0.114	0.246	1.000	26→0.124 27→0.055	0.231	1.000	52→0.0 53→0.0

2.2. Relative tests

2.2.1. GAHMDI

This test (Toreti *et al.*, 2012) aims to find the K possible change points and the homogeneous segments using Hidden Markov models. It needs the standardized difference of the candidate and the reference series for temperature variable. To avoid convergence to a local maximum, it uses a GA. The procedure starts by deciding on the maximum number of change points, $K_{\max}$. It uses a likelihood approach and the maximization is done by an expectation-maximization procedure. To get global maxima, GA is used to estimate the initial state sequence. The number of states is found through the MDL principle.

2.2.2. The Bayesian technique of change-point analysis

This is also known as the product partition model (Barry and Hartigan, 1992, 1993). It can be used as an absolute or a relative test. It presents the posterior probability of a break point for all time points of the series, and the approximate posterior mean through Markov Chain Monte Carlo (MCMC) method. As the unknown change points are treated as random variables, this method is also suitable in the identification of inhomogeneities. For the detection of multiple change-points in the mean of normally distributed variables, Barry and Hartigan (1993) assume the change-points to be identical, having the geometric distribution and independent priors for the parameters. Later, the prior distribution can be used as a normal distribution. An MCMC approximation of Barry and Hartigan's Bayes procedure is chosen to shorten the calculation time. The algorithm uses a partition $U=(U_1, U_2, \cdots, U_n)$ where $U_i=1$ indicates a change point at position $i+1$. At the beginning, U_i is equal to 0 for all $i<n$, with $U_n=1$. In each step of the Markov chain, at each position i, a value of U_i is drawn from the conditional distribution of U_i given all observations and the current partition. The transition probability for the conditional probability of a change point at the position $i+1$ may be obtained from the ratio

$$\frac{p_i}{1-p_i}=\frac{P\left(U_i=1\left|\mathbf{X},\ U_j,\ j\neq i\right.\right)}{P\left(U_i=0\left|\mathbf{X},\ U_j,\ j\neq i\right.\right)}$$

where $\boldsymbol{X}$ is the series. The series $\boldsymbol{X}$ can be only $\boldsymbol{Y}$ or it may also include some explanatory variables, such as reference information. If there is a mean shift in the series, posterior probability takes its maximum value. Calculations can be handled using the R package bcp (R-development Core Team, 2013), which is implemented by Erdman and

© 2015 Royal Meteorological Society

10970088, 2016, 1, Downloaded from https://rmets.onlinelibrary.wiley.com/doi/10.1002/joc.4329 by University Of Texas At, Wiley Online Library on [16/10/2024]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Table 9. Percentage of inhomogeneous results when the mean shift of two unit increase is used at (1) the beginning of the series (break starts at year 5), (2) the middle of the series (break starts at year 27), and (3) the end of the series (break starts at year 53) when there is a trend in the series.

Test	Inhomogeneity detection (%) (1)		Breakpoint detection (%)	Inhomogeneity detection (%) (2)		Breakpoint detection (%)	Inhomogeneity detection (%) (3)		Breakpoint detection (%)
	Y_t	Y_t, shift		Y_t	Y_t, shift		Y_t	Y_t, shift	
SNHT	0.054	0.723	4→0.411 5→0.103	0.050	0.999	26→0.465 27→0.123	0.050	0.956	52→0.427 53→0.111
Pettitt	0.096	0.161		0.088	0.708	26→0.130 27→0.082	0.089	0.278	52→0.0
Buishand range	0.049	0.069	4→0.002 5→0.003	0.051	0.463	26→0.182 27→0.113	0.050	0.117	52→0.032
F-test	0.055	0.115	–	0.048	0.569	26→0.137 27→0.084	0.047	0.245	51→0.218
ADF	0.247	0.227	–	0.253	0.330	–	0.251	0.354	–
KPSS	0.048	0.114	–	0.049	0.613	–	0.043	0.206	–
Von Neumann	0.949	0.923	–	0.950	0.804	–	0.946	0.888	–
Kruskal–Wallis	0.044	0.046	–	0.040	0.266	–	0.039	0.152	–
Friedman	0.040	0.042	–	0.040	0.220	–	0.039	0.110	–
F-test with references	0.059	0.368	–	0.059	0.999	26→0.451 27→0.125	0.064	0.859	51→0.471
Bayes	–	–	4→0.0 5→0.0	–	–	26→0.003 27→0.003	–	–	52→0.0 53→0.0
Bayes with reference	–	–	4→0.0 5→0.0	–	–	26→0.0 27→0.0	–	–	52→0.0 53→0.0
TPR	0.061	0.061	4→0.148 5→0.076	0.062	0.062	26→0.310 27→0.189	0.066	0.066	52→0.293 53→0.169
CauMe	0.319	0.459	4→0.114 5→0.058	0.308	0.753	26→0.080 27→0.046	0.301	0.569	52→0.080 53→0.080
RHTest	0.093	1.000	4→0.193 5→0.200	0.099	1.000	26→0.097 27→0.070	0.096	1.000	52→0.0 53→0.0
GAHMDI	0.237	0.300	4→0.073 5→0.266	0.250	0.301	26→0.096 27→0.024	0.242	0.305	52→0.0 53→0.0

Emerson (2007). The posterior probability plot gives information about the location of the possible shifts where the plot reaches its maximum. There could be several peaks representing several change points existing within the series.

2.2.3. F-*Test for structural breaks (or Chow test)*

This test can be used as a relative test or an absolute test. In this test, the locations of the break points or abrupt shifts in a time series are the unknown variables estimated by ordinary least squares. Consider a linear regression model, for which we assume that a break in one or more of the parameters occurred in period K,

$$Y_t = X_t'\beta_t + \varepsilon_t, \quad t = 1, 2, \cdots, n$$

where $X_t = (1, X_1, X_2, \cdots, X_r)$ is a $(r+1)\times 1$ vector of observations of the independent variables, ε_ts are independent and identically distributed random variables with mean *0* and constant variance, and β_t is the $(r+1)\times 1$ vector of regression coefficients. When X_t is the matrix with the observations from r reference stations, the test can be used as a relative test, but if $r = 0$, then it turns out to be an absolute test. Tests on structural change are actually testing the structural stability, i.e. the absence of structural change within the series. Hence, the null hypothesis is $H_0 : \beta_t = \beta_0$. However, the alternative hypothesis states the existence of a structural break in the series. Thus, in the alternative hypothesis, the coefficients are defined as

$$\beta_t = \begin{cases} \beta_A, 1 \le t \le K \\ \beta_B, K < t < n \end{cases}$$

where K is the change point in the interval $(k,\ n-k)$. Using an F-test and critical values of the F distribution, a test can be conducted. One drawback of the Chow test is that the change point, K has to be known in advance. Hence, the extension of the Chow test can be used when the change point is unknown. In this case, the F-test statistics for all potential change points have to be computed within a given time interval $(1,\ n)$. Calculations can be handled using the R package strucchange (R-development Core Team, 2013), which is implemented by Zeileis *et al.* (2003).

2.2.4. *SNH test*

This parametric method (Alexandersson, 1986) is similar to the likelihood ratio test and is based on the ordered values of observations under the independence and normality assumption. It uses standardized values. The test statistic is given by

$$T(m) = m\bar{z}_1^2 + (n-m)\,\bar{z}_2^2$$

© 2015 Royal Meteorological Society

10970088, 2016, 1, Downloaded from https://rmets.onlinelibrary.wiley.com/doi/10.1002/joc.4329 by University Of Texas At, Wiley Online Library on [16/10/2024]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Table 10. Percentage of inhomogeneous results when there are multiple mean shifts of 0.5 unit increase starting from 10th, 1 unit increase starting from 30th, and 2 unit increase starting from 50th year.

Test	Inhomogeneity detection (%)		Breakpoint detection (%)	TEST	Inhomogeneity detection (%)		Breakpoint detection (%)
	Y_t	Y_t, shift			Y_t	Y_t, shift	
SNHT	0.047	0.995	10→0.0 30→0.014 50→0.047	Friedman	0.038	0.161	–
Pettitt	0.090	0.498	10→0.0 30→0.030 50→0.002	F-test with references	0.060	0.975	10→0.001 30→0.020 50→0.090
Buishand range	0.051	0.208	10→0.0 30→0.048 50→0.006	Bayes	–	–	10→0.0 30→0.0 50→0.0
F-test	0.051	0.426	10→0.011 30→0.020 50→0.070	Bayes with reference	–	–	10→0.0 30→0.0 50→0.0
ADF	0.245	0.311	–	TPR	0.060	0.060	10→0.016 30→0.028 50→0.618
KPSS	0.046	0.448	–	CauMe	0.308	0.684	10→0.010 30→0.014 50→0.026
Von Neumann	0.945	0.837	–	RHTest	0.095	0.985	10→0.016 30→0.071 50→0.088
Kruskal–Wallis	0.042	0.220	–	GAHMDI	0.251	0.987	10→0.015 30→0.080 50→0.117

where $m = 1, \ldots, n$ and

$$\bar{z}_1 = \frac{1}{m}\sum_{i=1}^{m}\frac{\left(Q_i - \overline{Q}\right)}{s} \text{ and}$$

$$\bar{z}_2 = \frac{1}{n-m}\sum_{i=m+1}^{n}\frac{\left(Q_i - \overline{Q}\right)}{s} \text{ and}$$

$$Q_i = Y_i - \sum_{j=1}^{r}\rho_j^2\left[X_{ji} - \overline{X}_j - \overline{Y}\right] \Big/ \sum_{j=1}^{r}\rho_j^2$$

here, X_{ji}s shows the reference series and ρ_j represents the correlation between the candidate series and reference series. The two reference stations are used to calculate Q_i term, which is similar to calculate weighted average where weights are the squared correlation coefficients between the reference station and the test station. Thus, the ρ_j puts more weight to the highly correlated series (Tuomenvirta, 2002). When a mean shift occurs, $T(m)$ takes its maximum value. Therefore, the test statistic is the global maximum of $T(m)$. Moreover, in the simulation studies of Ducre-Robitaille *et al.* (2003), DeGaetano (2006) and Toreti *et al.* (2011), the false detection rate of the SNHT increases at the beginning and the end of the series, even though Wijngaard *et al.* (2003) finds the reverse results in their simulation study. Aguilar *et al.* (2003) suggest to use the SNHT (or other homogeneity detection tests) after a standardized difference of the temperature series for the candidate and reference series. These conflicting results might be the result of misuse of the SNHT technique. Although it is sensitive to a single shift in the mean, it cannot find more than one properly. In that case, the single shift test on two or more consecutive parts of a complicated series can be used (Alexandersson, 1986; Alexandersson and Moberg, 1997; Moberg and Alexandersson, 1997). This test needs a homogeneous reference series, which has to be highly correlated with the series tested. The test assumes independency in the aggregated time series dataset after obtaining a standardized series. Even though for temperature series, using annually aggregated series may eliminate or reduce the dependency, there might still be autocorrelation within the pre-processed datasets used in the test statistic. Even though SNHT has these drawbacks, it is the most popular inhomogeneity test and it is a good test to detect a single mean shift (Yazici *et al.* 2012).

2.2.5. *RHTest*

Wang, 2008a, 2008b proposes an empirical approach to account for lag-1 autocorrelation to detect single or multiple mean shifts in white or red (first-order autoregressive) time series. The method uses penalized maximal t-test (PMT) or F-test (PMF) proposed by (Wang *et al.*, 2007). In this method, the most probable change point c_0 ($t \in \{n_{\min}, n_{\min}+1, \ldots, n-n_{\min}\}$) is identified. Then, the next most probable change points; c_{01} and c_{03} are sought out from the segments $t \in \{n_{\min}, \ldots, (c_0 - n_{\min})\}$ and $t \in \{c_0 + 1 + n_{\min}, \ldots (n - n_{\min})\}$, respectively. Then,

© 2015 Royal Meteorological Society

10970088, 2016, 1, Downloaded from https://rmets.onlinelibrary.wiley.com/doi/10.1002/joc.4329 by University Of Texas At, Wiley Online Library on [16/10/2024]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Table 11. Percentage of inhomogeneous results when the mean shift of 0.5 unit increase is used at one of the reference series at (1) the beginning of the series (break starts at year 5), (2) the middle of the series (break starts at year 27),and (3) the end of the series (break starts at year 53).

Test	Inhomogeneity detection (%) (1)		Breakpoint detection (%)	Inhomogeneity detection (%) (2)		Breakpoint detection (%)	Inhomogeneity detection (%) (3)		Breakpoint detection (%)
	Y_t	Y_t, shift		Y_t	Y_t, shift		Y_t	Y_t, shift	
SNHT	0.048	0.054	4→0.042 5→0.033	0.048	0.078	26→0.030 27→0.026	0.049	0.062	52→0.040 53→0.041
F-test with references	0.057	0.060	–	0.060	0.076	26→0.045 27→0.022	0.061	0.067	51→0.066
Bayes with reference	–	–	4→0.0 5→0.0	–	–	26→0.0 27→0.0	–	–	52→0.0 53→0.0
TPR	0.062	0.062	4→0.011 5→0.004	0.064	0.064	26→0.023 27→0.026	0.061	0.061	52→0.050 53→0.047
CauMe	0.303	0.312	4→0.037 5→0.036	0.304	0.307	26→0.009 27→0.012	0.313	0.316	52→0.013 53→0.012
RHTest	0.098	0.114	4→0.033 5→0.034	0.103	0.230	26→0.057 27→0.047	0.097	0.141	52→0.050 53→0.036
GAHMDI	0.242	0.277	4→0.077 5→0.088	0.245	0.244	26→0.005 27→0.0	0.235	0.234	52→0.0 53→0.0

Table 12. Percentage of inhomogeneous results when the mean shift of 1 unit increase is used at one of the reference series at (1) the beginning of the series (break starts at year 5), (2) the middle of the series (break starts at year 27), and (3) the end of the series (break starts at year 53).

Test	Inhomogeneity detection (%) (1)		Breakpoint detection (%)	Inhomogeneity detection (%) (2)		Breakpoint detection (%)	Inhomogeneity detection (%) (3)		Breakpoint detection (%)
	Y_t	Y_t, shift		Y_t	Y_t, shift		Y_t	Y_t, shift	
SNHT	0.047	0.065	4→0.085 5→0.035	0.048	0.158	26→0.064 27→0.046	0.045	0.090	52→0.072 53→0.066
F-test with references	0.060	0.063	–	0.058	0.126	26→0.053 27→0.045	0.059	0.078	51→0.100
Bayes with reference	–	–	4→0.0 5→0.0	–	–	26→0.0 27→0.0	–	–	52→0.0 53→0.0
TPR	0.063	0.063	4→0.026 5→0.022	0.062	0.062	26→0.073 27→0.040	0.064	0.064	52→0.010 53→0.046
CauMe	0.324	0.306	4→0.040 5→0.028	0.308	0.310	26→0.008 27→0.009	0.313	0.311	52→0.012 53→0.016
RHTest	0.095	0.172	4→0.067 5→0.093	0.085	0.586	26→0.098 27→0.010	0.091	0.298	52→0.113 53→0.091
GAHMDI	0.248	0.379	4→0.113 5→0.170	0.252	0.684	26→0.132 27→0.016	0.244	0.430	52→0.115 53→0.076

the new estimate of $c_{0,}$ represented by c_{02} is identified. $PT_{\max}$ test statistics is needed to determine the most significant change point. To estimate the *p*-value of the $PT_{\max}$, the following models are used

$$R_t = X_t - \widehat{X}_t, \quad t = 1, 2, \ldots, n$$
$$W_1 = R_1$$
$$W_t = R_t - \widehat{\phi} R_{t-1}, \quad t = 2, \ldots, n$$

where $\widehat{X}_t$ denotes the full model fit to the series X_t, and R_t denotes the residual series. The prewhitened series, W_t is used to calculate the *t*-statistics and *p*-value for *c*. The change point with the maximum $PT_{\max}$ statistic is identified. If it is found significant, this point is taken as the identified change point.

2.2.6. *CauMe (HOMER)*

Caussinus and Mestre (2004) propose this method to detect the change points and correct them in a climate series. In this method, it is assumed that the distribution of the difference between the test and reference stations is normal. The detection procedure is based on an adapted penalized log-likelihood procedure and can be used even if there is no reliable reference series. It is assumed that each observation in a climate series reflects the climate effect, station effect and a random white noise. The comparison with the reference stations is made by the following two-factor model

$$E\left(X_{ij}\right) = \mu_i + \nu_{jh(i,j)}$$

where μ_i is the climate effect at time *i* and ν_{jh} is the station effect of station *j*. The number of hypotheses increases if

© 2015 Royal Meteorological Society

10970088, 2016, 1, Downloaded from https://rmets.onlinelibrary.wiley.com/doi/10.1002/joc.4329 by University Of Texas At, Wiley Online Library on [16/10/2024]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Table 13. Percentage of inhomogeneous results when the mean shift of 2 unit increase is used at one of the reference series at (1) the beginning of the series (break starts at year 5), (2) the middle of the series (break starts at year 27), and (3) the end of the series (break starts at year 53).

Test	Inhomogeneity detection (%) (1)		Breakpoint detection (%)	Inhomogeneity detection (%) (2)		Breakpoint detection (%)	Inhomogeneity detection (%) (3)		Breakpoint detection (%)
	Y_t	Y_t, shift		Y_t	Y_t, shift		Y_t	Y_t, shift	
SNHT	0.052	0.132	4→0.204 5→0.103	0.051	0.536	26→0.168 27→0.090	0.050	0.254	52→0.178 53→0.102
F-test with references	0.064	0.089	–	0.061	0.315	26→0.134 27→0.070	0.060	0.148	50→0.084
Bayes with reference	–	–	4→0.0 5→0.0	–	–	26→0.0 27→0.0	–	–	52→0.0 53→0.0
TPR	0.065	0.066	4→0.127 5→0.087	0.060	0.060	26→0.378 27→0.378	0.063	0.063	52→0.346 53→0.171
CauMe	0.308	0.308	4→0.041 5→0.026	0.300	0.310	26→0.006 27→0.011	0.311	0.307	52→0.013 53→0.013
RHTest	0.097	0.446	4→0.233 5→0.195	0.091	0.979	26→0.023 27→0.193	0.097	0.779	52→0.210 53→0.200
GAHMDI	0.234	0.650	4→0.154 5→0.417	0.240	0.977	26→0.324 27→0.105	0.250	0.805	52→0.320 53→0.108

Table 14. Percentage of inhomogeneous results when the sudden decrease is used at one of the reference series at (1) the beginning of the series, (2) the middle of the series, and (3) the end of the series.

Test	Inhomogeneity detection (%) (1)		Breakpoint detection (%)	Inhomogeneity detection (%) (2)		Breakpoint detection (%)	Inhomogeneity detection (%) (3)		Breakpoint detection (%)
	$Y_{t,}$	Y_t, shift		$Y_{t,}$	Y_t, shift		Y_t	Y_t, shift	
SNHT	0.051	0.050	4→0.027 5→0.033	0.047	0.048	26→0.008 27→0.002	0.051	0.052	52→0.030 53→0.024
F-test with references	0.061	0.062	–	0.062	0.063	26→0.023 27→0.020	0.066	0.067	–
Bayes with reference	–	–	4→0.0 5→0.0	–	–	26→0.0 27→0.0	–	–	52→0.0 53→0.0
TPR	0.062	0.062	4→0.006 5→0.0	0.065	0.065	26→0.013 27→0.023	0.061	0.061	52→0.030 53→0.022
CauMe	0.308	0.314	4→0.035 5→0.035	0.320	0.313	26→0.010 27→0.009	0.314	0.315	52→0.012 53→0.013
RHTest	0.094	0.094	4→0.021 5→0.023	0.093	0.092	26→0.026 27→0.010	0.090	0.093	52→0.030 53→0.015
GAHMDI	0.250	0.247	4→0.060 5→0.050	0.241	0.234	26→0.013 27→0.011	0.247	0.248	52→0.013 53→0.021

the length of the series is large. To make it computationally feasible, pairwise comparisons are made for preselection by using a dynamic programming algorithm.

2.2.7. *TPR*

This method is proposed by Easterling and Peterson (1995). A series of differences between the candidate series and the reference series is first computed, as in the SNHT approach. A simple regression model (SLR), where the series of differences is the dependent variable and time is the independent variable, is fitted to the entire series and the residual sum of squares RSS_1 is obtained. Then, two separate SLR models are constructed before and after the potential step v and the sum of the residual sum of squares (RSS_2) of these two models are calculated for each potential break v. The maximum RSS_2 is used to calculate the test statistic. The statistical significance of the TPR is then tested using the statistic U:

$$U = \frac{\left(\mathrm{RSS}_1 - \mathrm{RSS}_2\right)/3}{\mathrm{RSS}_2/(n-4)} \sim F_{3,n-4}$$

The critical values can be found in the study by Lund and Reeves (2002). If the test indicates a potential step at the position v, then the magnitude of the shift is estimated by the difference between the averages before and after the step.

© 2015 Royal Meteorological Society

10970088, 2016, 1, Downloaded from https://rmets.onlinelibrary.wiley.com/doi/10.1002/joc.4329 by University Of Texas At, Wiley Online Library on [16/10/2024]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Table 15. Percentage of inhomogeneous results when the gradual increase is used at at one of the reference series (1) the beginning of the series (between 5th and 10th realizations), (2) the middle of the series (between 27th and 32nd realizations), and (3) the end of the series (between 53rd and 58th realizations).

Test	Inhomogeneity detection (%) (1)		Breakpoint detection (%)	Inhomogeneity detection (%) (2)		Breakpoint detection (%)	Inhomogeneity detection (%) (3)		Breakpoint detection (%)
	Y_t	Y_t, shift		Y_t	Y_t, shift		Y_t	Y_t, shift	
SNHT	0.050	0.050	4→0.021 5→0.011	0.049	0.049	26→0.010 27→0.004	0.046	0.046	52→0.034 53→0.021
F-test with references	0.063	0.065	4→0.0 5→0.0	0.064	0.065	26→0.009 27→0.021	0.061	0.061	51→0.057
Bayes with reference	–	–	4→0.0 5→0.0	–	–	26→0.0 32→0.0	–	–	52→0.0 53→0.0
TPR	0.062	0.062	4→0.003 5→0.004	0.063	0.063	26→0.012 32→0.018	0.061	0.061	52→0.043 53→0.035
CauMe	0.310	0.310	4→0.036 5→0.031	0.305	0.305	26→0.012 32→0.009	0.312	0.312	52→0.011 53→0.012
RHTest	0.087	0.102	4→0.037 5→0.017	0.097	0.100	26→0.017 27→0.015	0.093	0.098	52→0.018 53→0.026
GAHMDI	0.232	0.250	4→0.061 5→0.041	0.245	0.247	26→0.0 27→0.0	0.237	0.240	52→0.0 53→0.0

Table 16. Percentage of inhomogeneous results when the increasing trend with slope (1) 0.00075 (monthly) and (2) 0.0033 (monthly) are applied to one of the reference series.

Test	Inhomogeneity detection (%) (1)		Inhomogeneity detection (%) (2)	
	Y_t	Y_t, shift	Y_t	Y_t, shift
SNHT	0.048	0.067	0.046	0.344
F-test with references	0.063	0.075	0.058	0.227
Bayes with reference	–	–	–	–
TPR	0.064	0.064	0.058	0.058
CauMe	0.309	0.308	0.305	0.307
RHTest	0.097	1.000	0.092	1.000
GAHMDI	0.241	0.999	0.244	0.236

3. Simulation study

3.1. Data generation

Before the simulation study, we analysed the time series structure of the monthly average temperature of 25 meteorological stations in Turkey from 1950 to 2006 using SAS 9.2 Time Series Forecasting System (SAS Institute Inc., 2008). It can model multiple time series automatically and finds the best fitting model for the series using minimum RMSE criterion. For all the temperature series, the software suggests the seasonal exponential smoothing method that best fits these series with different parameter estimates. Then, we chose one station haphazardly and used its parameter estimates to generate a temperature series. The selected station was Zonguldak station. Hence, monthly temperature series were generated using the following model with seasonal exponential smoothing equation

$$Y_t = \mu_t + S_p(t) + \varepsilon_t \quad (1)$$

where μ_t is a level parameter representing the mean of the series, $S_p(t)$s are seasonal parameters and ε_t is an error term $t = 1, 2, \cdots, 720; p =$ seasonality period $= 12$. Here we assume that ε_t is normally distributed with 0 mean and a constant variance.

Using the best fitted model for the temperature series of Zonguldak station, we used $\mu_t = 16$ and the 12 seasonal parameters as

$$S_{12}(1) = -7.4, \quad S_{12}(2) = -7.5, \quad S_{12}(3) = -6.1,$$
$$S_{12}(4) = -3.0, \quad S_{12}(5) = 1.4, \quad S_{12}(6) = 6.2$$
$$S_{12}(7) = 9.0, \quad S_{12}(8) = 9.4, \quad S_{12}(9) = 5.7,$$
$$S_{12}(10) = 1.8, \quad S_{12}(11) = -2.6 \text{ and } S_{12}(12) = -5.5$$

and these factors repeat themselves year to year. We generated 720 time points representing monthly temperature (60 years). We also added a random error term, which is explained below, to the generated series. Then, yearly series were obtained by aggregating the monthly values. After generating data, we applied a paired t-test to the monthly and yearly generated temperature series and temperature series of the Zonguldak station because we generate data using a normal distribution. The two test results did not find statistically significant differences among the generated and original series (p-value for monthly series $= 0.9338 > 0.05$ and p-value for yearly series $= 0.9165 > 0.05$). The time series plots of the monthly series (not given here) also suggest a good fit. Hence, it can be said that the generated series captured the behaviour of the original series well and could be used to represent temperature series.

© 2015 Royal Meteorological Society

10970088, 2016, 1, Downloaded from https://rmets.onlinelibrary.wiley.com/doi/10.1002/joc.4329 by University Of Texas At, Wiley Online Library on [16/10/2024]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Table 17. Percentage of inhomogeneous results when the mean shift of 0.5 unit increase is used at one of the reference series (1) the beginning of the series (break starts at year 5), (2) the middle of the series (break starts at year 27), and (3) the end of the series (break starts at year 53) when there is a trend in the series.

Test	Inhomogeneity detection (%) (1)		Breakpoint detection (%)	Inhomogeneity detection (%) (2)		Breakpoint detection (%)	Inhomogeneity detection (%) (3)		Breakpoint detection (%)
	Y_t	Y_t, shift		Y_t	Y_t, shift		Y_t	Y_t, shift	
SNHT	0.048	0.095	4→0.014 5→0.008	0.051	0.538	26→0.030 27→0.024	0.047	0.436	52→0.025 53→0.023
F-test with references	0.061	0.089	–	0.062	0.326	26→0.076 27→0.040	0.058	0.260	51→0.050
Bayes with reference	–	–	4→0.0 5→0.0	–	–	26→0.0 27→0.0	–	–	52→0.0 53→0.0
TPR	0.060	0.060	4→0.008 5→0.013	0.065	0.065	26→0.030 27→0.027	0.068	0.068	52→0.030 53→0.040
CauMe	0.308	0.308	4→0.035 5→0.031	0.306	0.306	26→0.008 27→0.011	0.312	0.312	52→0.014 53→0.014
RHTest	0.086	1.000	4→0.037 5→0.047	0.089	1.000	26→0.042 27→0.038	0.095	1.000	52→0.0 53→0.0
GAHMDI	0.244	0.999	4→0.024 5→0.037	0.241	1.000	26→0.066 27→0.042	0.251	0.999	52→0.0 53→0.0

For SNHT, *F*-test with reference test, Bayesian with reference test, TPR, CauMe, GAHMDI and RHTest, we needed to generate a highly correlated reference series. Because of this, we also generated two reference series by changing the given parameters in model (1) with similar numeric values. However, this does not help to obtain a highly correlated annual series with the original one for the annual series. Therefore, the error terms ε_t are generated from

$$\varepsilon_t \sim N_3\left(\mu = \begin{bmatrix}0\\0\\0\end{bmatrix}, \quad \sum = \begin{bmatrix}1 & .9 & .9\\ .9 & 1 & .9\\ .9 & .9 & 1\end{bmatrix}\right)$$

In this way, the correlation between the original (or candidate) and reference series is above 85% in the annual basis.

Using the definition of inhomogeneity in climate studies, several scenarios were considered as the incidence of the inhomogeneity on the annual series to account for the observed inhomogeneity: mean shift to represent abrupt discontinuities, gradual change and sudden change to represent the changes in the surroundings of the station or changes in the instrumental attributes or maintenance/calibration problems related to the apparatus, and trend to represent the change, such as urbanization around the station. Hence, we applied the mean shift starting at the beginning, in the middle and at the end of the series (in the 60 years, the breaks are located at the 5th, 27th and 53rd year, respectively) for the change in the location of the station using 0.5, 1 and 2 °C shifts; gradual change case again considered at the beginning, in the middle and at the end of the series; sudden decrease (1 °C from the yearly aggregates) at one time point for the same three different locations on the series; and trend. A gradual change was created by adding $1/(1+i), i = 1, 2, \cdots, 6$ to the observations at the beginning, in the middle and at the end of the series giving the effect lasting for a period of 6 years which was chosen haphazardly (hence, a sudden increase of 0.50 °C was gradually decreased to 0.14 °C and died out); and lastly a trend was formed as $Y_t + \alpha t; \alpha = 0.01, 0.04; t = 1, 2, \cdots, 60$. We also considered the mean shift where there is a climate change effect on the series. A slow increasing trend representing the global warming was added to the generated series as $Y_t + 0.001t$, $t = 1, 2, \cdots, 60$ and then mean shifts of amounts 0.5, 1 and 2 °C were added at the beginning, in the middle and at the end of the series. For the performance of the multiple change point detection methods, the mean of the candidate series was changed after the 10th (30th and 50th) value; this is because the average length of the periods between breaks is about 20 years (Lindau and Venema, 2013).

RHTest runs on monthly series, so the scenarios were adapted to monthly data. The mean shift was applied on all months after the starting point using 0.5/12,1/12 and 2/12 °C shifts; gradual change case was again considered at the beginning, in the middle and at the end of the series; sudden decrease (1 °C from any month) at one time point for the same three different locations on the series; and trend. A gradual change was created by adding $1/(18+i)$, $i = 1, 2, \ldots, 72$ to the observations at the beginning, in the middle and at the end of the series, to have the similar gradual effect on the annual series. Lastly a trend was formed as $Y_t + \alpha t$; $\alpha = 0.00075, 0.0033; t = 1, 2, \cdots, 720$. We also considered the mean shift where there was a climate change effect on the series. A slow increasing trend was added to the generated series as $Y_t + 0.00075t$, $t = 1, 2, \cdots, 720$ and then mean shifts of amounts 0.5/12, 1/12 and 2/12 °C were added at the beginning, in the middle and at the end of the series. For the performance of the multiple change point detection methods, the mean of the candidate series was changed after the 120th (360th and 600th) value.

© 2015 Royal Meteorological Society

10970088, 2016, 1, Downloaded from https://rmets.onlinelibrary.wiley.com/doi/10.1002/joc.4329 by University Of Texas At, Wiley Online Library on [16/10/2024]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Table 18. Percentage of inhomogeneous results when the mean shift of 1 unit increase is used at at one of the reference series (1) the beginning of the series (break starts at year 5), (2) the middle of the series (break starts at year 27), and (3) the end of the series (break starts at year 53) when there is a trend in the series.

Test	Inhomogeneity detection (%) (1)		Breakpoint detection (%)	Inhomogeneity detection (%) (2)		Breakpoint detection (%)	Inhomogeneity detection (%) (3)		Breakpoint detection (%)
	Y_t	Y_t, shift		Y_t	Y_t, shift		Y_t	Y_t, shift	
SNHT	0.048	0.095	4→0.168 5→0.112	0.048	0.067	26→0.005 32→0.016	0.050	0.155	52→0.078 53→0.050
F-test with references	0.061	0.089	–	0.062	0.071	26→0.015 32→0.020	0.061	0.120	51→0.112
Bayes with reference	–	–	4→0.0 5→0.0	–	–	26→0.0 32→0.0	–	–	52→0.0 53→0.0
TPR	0.061	0.061	4→0.020 5→0.014	0.063	0.063	26→0.080 32→0.071	0.063	0.063	52→0.095 53→0.064
CauMe	0.305	0.304	4→0.038 5→0.033	0.308	0.306	26→0.010 32→0.009	0.310	0.318	52→0.008 53→0.014
RHTest	0.097	1.000	4→0.076 5→0.086	0.097	1.000	26→0.065 27→0.050	0.099	1.000	52→0.0 53→0.0
GAHMDI	0.238	0.300	4→0.005 5→0.124	0.243	0.307	26→0.057 27→0.027	0.252	0.999	52→0.0 53→0.0

Table 19. Percentage of inhomogeneous results when the mean shift of 2 unit increase is used at one of the reference series (1) the beginning of the series (break starts at year 5); (2) the middle of the series (break starts at year 27); (3) the end of the series (break starts at year 53) when there is a trend in the series.

Test	Inhomogeneity detection (%) (1)		Breakpoint detection (%)	Inhomogeneity detection (%) (2)		Breakpoint detection (%)	Inhomogeneity detection (%) (3)		Breakpoint detection (%)
	Y_t	Y_t, shift		Y_t	Y_t, shift		Y_t	Y_t, shift	
SNHT	0.049	0.206	4→0.174 5→0.083	0.051	0.536	26→0.164 32→0.086	0.053	0.373	52→0.166 53→0.090
F-test with references	0.057	0.114	–	0.065	0.317	26→0.142 27→0.074	0.055	0.207	51→0.162
Bayes with reference	–	–	4→0.0 5→0.0	–	–	26→0.0 27→0.0	–	–	52→0.0 53→0.0
TPR	0.061	0.061	4→0.132 5→0.053	0.063	0.063	26→0.312 27→0.170	0.061	0.062	52→0.345 53→0.166
CauMe	0.317	0.314	4→0.038 5→0.024	0.306	0.306	26→0.013 27→0.013	0.315	0.306	52→0.014 53→0.011
RHTest	0.098	1.000	4→0.185 5→0.195	0.094	1.000	26→0.096 27→0.078	0.090	1.000	52→0.0 53→0.0
GAHMDI	0.240	0.998	4→0.062 5→0.220	0.252	1.000	26→0.266 27→0.065	0.242	0.999	52→0.0 53→0.0

The data generated 10 000 times under all scenarios were applied to all methods expect for GAHMDI and RHTest, for which data were simulated only 5000 times due to computational issues. Then, we counted how many times these tests detected the artificially created inhomogeneity and how many times the tests found inhomogeneity in the series that were homogeneous. We also checked the location of the mean shift for the inhomogeneous data. The ADF, KPSS, KW and the Friedman tests cannot give the location of the exact mean shift. In the application of the KW test, we used 10-year periods. We also used a 10-year period in the Friedman test to compare the results of two tests. For the Bayesian test, we stored the location of the maximum point(s). Considering no explanatory variables, the test was used as an absolute test (in our study, it was named as Bayes). It was also used as relative test (named as Bayes with reference). We applied the *F*-test for structural breaks as an absolute test without an independent variable but with a constant term in the model (named as *F*-test), and as a relative test using the reference series as independent variables (named as *F*-test with reference).

3.2. Results when both reference series are homogeneous

The simulation results given in Tables 1–10 show the performances of all the scenarios covered in the study when both reference series are homogeneous. Subtables *i*, *ii* and *iii* also include three types of inhomogeneities, which were created at the beginning, in the middle and at the end of the series (in the 60 years, the breaks start in the 5th, 27th and 53rd year, respectively). The column named as 'Y_t' shows the inhomogeneity detection rate among the 10 000 simulations before creating a breakpoint. The column named as 'Y_t, *shift*' shows the inhomogeneity rate among the 10 000 simulations after creating a change

© 2015 Royal Meteorological Society

10970088, 2016, 1, Downloaded from https://rmets.onlinelibrary.wiley.com/doi/10.1002/joc.4329 by University Of Texas At, Wiley Online Library on [16/10/2024]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Table 20. Percentage of inhomogeneous results when there are multiple mean shifts of 0.5 unit increase starting from 10th, 1 unit increase starting from 30th, and 2 unit increase starting from 50th year at one of the reference series

Test	Inhomogeneity detection (%) (1)		Breakpoint detection (%)
	Y_t	Y_t, shift	
SNHT	0.048	0.095	10→0.026 30→0.065 50→0.010
F-test with references	0.061	0.089	10→0.022 30→0.014 50→0.018
Bayes with reference	–	–	10→0.0 30→0.0 50→0.0
TPR	0.062	0.062	10→0.016 30→0.033 50→0.553
CauMe	0.310	0.306	10→0.015 30→0.011 50→0.010
RHTest	0.093	0.956	10→0.008 30→0.026 50→0.021
GAHMDI	0.240	0.299	10→0.043 30→0.064 50→0.005

point. The break point detection column shows where the test identified the inhomogeneity and its detection rate among the inhomogeneous test results. Empty cells in this column denote cases in which the test cannot give breakpoint information.

It was observed that the best tests for detecting inhomogeneities were SNHT, RHTest, F-test with reference, GAHMDI and F-test. All these tests found the mean shift very well. When the shift was at the beginning of the series, the chance of detecting it was low, especially with F-test, even though F-test with references had a high detection rate when compared with the F-test. However, the RHTest which considers autocorrelation in the series detected the inhomogeneity well, even when it was at the beginning of the series. GAHMDI was the best method to detect the sudden decrease if the breakpoint is in the middle or at the end of the series, even it also detected a high percentage of inhomogeneity before creating a breakpoint. ADF and Von Neumann were also the better tests for detecting a sudden change in the series, but their inhomogeneity detection rates were also high before creating the break. Hence, they are not reliable. The tests did not identify the gradual change and trend type inhomogeneity within series very well. The detection rates were almost the same when there is breakpoint and without a breakpoint. The best test for detecting trend in the series was the RHTest. SNHT, F-test with references and CauMe follow the RHTest. The Bayes test cannot be used to search for the existence of a trend in the series. It is designed to find break points only by representing break points on graphics. The Von Neumann test gave good results, but only when we considered the case where we had created inhomogeneities. When we considered the original series, which are homogeneous, the Von Neumann test rejected the homogeneity at very high rates. Hence, it was not a reliable test. The worst tests were Friedman, KW and Pettitt tests. KW test gave better results than the Friedman test even though it is designed for independent data. The KW test found high mean shifts and sharp trends easily but failed to detect gradual changes. In addition, it even occasionally showed wrong points as a break point. Actually, the KW and Friedman tests gave only the location of the break in certain time interval. They did not present the exact break point. Consequently, the Friedman, KW, Von Neumann and Pettitt tests may not be reliable tests for investigating homogeneity, especially if there are no metadata. The Bayes tests, KPSS test and Buishand range test showed moderate performances as homogeneity tests. In the mean shift in addition to trend case, the detection rates were higher when compared with the only mean shift case. In both cases, the detection rates increased as the amount of shift increased. For this scenario, the RHTest detected the change point in all cases but it failed to detect the exact locations. In general, the probability of detecting the inhomogeneity increased if the magnitude of the mean shift increased, and the probability of detecting the true breakpoint increased as the amount of the shift increased. Moreover, the chance of detecting inhomogeneity was higher if the break started in the middle of the series rather than in the cases where the break started at the beginning or end of the series. In the sudden change case, as there was a decrease in only one year, the probabilities of detecting the inhomogeneity and the breakpoint was not as high as the probabilities in other scenarios. In some of the tests (SNHT, Pettitt, F-test, KPSS, Kruskal–Wallis, Pettitt and F-test with references), the probability of detecting the true change point for the gradual change scenario was lower if the breakpoint was in the middle of the series rather than in the cases where the breakpoints started at the beginning or end of the series, in general. In the trend case, if the trend level increased, the probability of detecting the inhomogeneity increased for all tests except Von Neumann.

When there were multiple change points in the series, inhomogeneity detection rates of some of the tests were low (see Table 10). The detection rate was higher in SNHT, F-test with references, GAHMDI and RHTest. Moreover, the detection of the location of the last breakpoint was higher compared to previous breaks. Only TPR presented good results to detect the inhomogeneity at the end of the series. The researchers may consider developing more reliable tests for the detection of multiple breaks in the series.

3.3. Investigating the performance of relative tests with an inhomogeneous reference series

Relative tests need reliable reference series. The reference series have to be highly correlated with the tested series and they have to be homogeneous, but it is not possible to know whether the reference series are homogeneous

© 2015 Royal Meteorological Society

10970088, 2016, 1, Downloaded from https://rmets.onlinelibrary.wiley.com/doi/10.1002/joc.4329 by University Of Texas At, Wiley Online Library on [16/10/2024]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

or not if there is no metadata. In addition, the series may be homogeneous in terms of temperature series but not for precipitation. Therefore, the previous studies on different variables do not give any information. In this part of the study, we demonstrate the performance of the relative tests, namely SNHT, *F*-test with references, Bayes with reference, TPR, CauMe, GAHMDI and RHTest. As candidate series, we used the homogeneous series and created inhomogeneities in one of the reference series. Then, we applied the relative tests using all the scenarios. If the tests are robust, then it is expected not to reject the hypothesis of homogeneity. The results are given in Tables 11–20.

In the mean shift scenario, the chance of detecting the inhomogeneity was smaller in *F*-test with references, TPR and SNHT, when compared with CauMe, GAHMDI and RHTest. Moreover, the probabilities of detecting the inhomogeneity increased as the magnitude of the shift increased for SNHT and *F*-test with references. In the sudden decrease case, there was a small difference between the tests in terms of the chance of detecting the inhomogeneity, except GAHMDI and CauMe whose inhomogeneity detection rates were high even if there were no breakpoint. Even so, the probability of detection was very low. The probability of true breakpoint detection was higher in *F*-test with references even if the break was not at the candidate station. The chance of detecting a gradual increase was lower in all tests, but higher in the RHTest. In the trend case, the detection of the inhomogeneity increased as the α level increased. Moreover, the detection chance was the highest for the RHTest and GAHMDI. In the trend and mean shift case the RHTest detected inhomogeneity in all cases, while the chance of detecting the locations was small. All these results show that if the series were found to be inhomogeneous, another series of tests had to be applied to be sure that reference series were indeed homogeneous. SNHT and F test with reference seem more robust to inhomogeneity in the reference series.

The results of detecting multiple change points in one of the reference stations are presented in Table 20. If the changes occurred in one of the reference stations, the detection rate of inhomogeneity for the test station was the highest in the RHTest. The detection chance was the same in TPR when there was no breakpoint and it was higher in SNHT when compared with the *F*-test with references. It can be said that TPR is the most robust test although the overall performance of the test was not good.

3.4. Comparison of the observed results with the literature

Ducre-Robitaille *et al.* (2003) concluded that SNHT no trend, MLR and Bayesian with reference produce the best solutions for the identification of homogeneous series. In their study, the detection of a single small step is better in SNHT and Bayesian with reference, while they did not perform well in our study. Domonkos (2008a) concluded that Easterling and Peterson's (1995) method, which is TPR, SNHT and MLR are the better tests, while TPR is not a satisfactory test in our study. Toreti *et al.* (2013) concluded that GAHMDI, SNHT and CauMe have similar performances in detecting single change points, while the RHTest underdetects it. In the multiple change points case, our results were similar, whereas GAHMDI performed better when a stepwise function with three change points were added. Guijarro (2013) indicated that classical *t*-test and SNHT perform better when there is a single break in the middle of the series. However, it should be noted that in our study, the generated data represent all time series features of real average temperature series, while most studies in the literature generate data from normal distribution without comparing the closeness of the generated series with the original temperature series.

4. Conclusion and further studies

Climate change is expected to have strong impact on our world. Analysing, modelling and forecasting meteorological variables are crucial to achieve a better understanding of the climate system and its changes. To be able to obtain reliable results, meteorological data have to be homogeneous. In this study, the data generation method originated from a real climate series and the performances of homogeneity tests were questioned. Even though they were not used as such in the literature, we suggest that ADF, KPSS and Friedman tests are potentially valid homogeneity tests.

Our simulation study shows that for the detection of inhomogeneties, one should consider the use of SNHT, *F*-test for structural breaks with reference series, GAHMDI and the RHTest. However, if there is no appropriate reference series, then the *F*-test for structural breaks test is the best options. The relative tests are substantially affected by inhomogeneties within the reference series. Therefore, they have to be used with at least one of the absolute tests. Our results also suggest that KW, Friedman, ADF, Von Neumann and Pettit tests may not be reliable for testing homogeneity, thus they should be used judiciously. After the detection of the inhomogeneity based on the tests, the series has to be homogenized.

As a further study, the performance of these tests under non-normality (especially skewed distributions for precipitation series) and high autocorrelation could be investigated. Moreover, other structural break tests which are created for time series data could be considered as a homogeneity test.

Acknowledgements

The authors would like to thank the anonymous reviewers for their excellent and constructive comments and suggestions. The authors would like to extend their thanks to all members of the NINLIL research group http://www.stat.metu.edu.tr/, for their support. The authors also would like to thank Gary Francis Conlan (School of Foreign Languages, METU, Turkey) for assessment of the language qualification of our manuscript and the editor and the anonyms referees for their valuable comments

© 2015 Royal Meteorological Society

10970088, 2016, 1, Downloaded from https://rmets.onlinelibrary.wiley.com/doi/10.1002/joc.4329 by University Of Texas At, Wiley Online Library on [16/10/2024]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

and criticisms. Their contributions lead to the improved version of this article.

References

Aguilar E, Auer I, Brunet M, Peterson TC, Wieringa J. 2003. *WCDMP No. 53, WMD/TD No. 1186: Guidelines on Climate Metadata and Homogenization*. World Meteorological Organization: Geneva, Switzerland.

Alexandersson H. 1986. A homogeneity test applied to precipitation data. *Int. J. Climatol.* **6**: 661–675.

Alexandersson H, Moberg A. 1997. Homogenization of Swedish temperature data, part I: homogeneity test for linear trends. *Int. J. Climatol.* **17**: 25–34.

Barry D, Hartigan JA. 1992. Product partition models for change point problem. *Ann. Stat.* **20**: 260–279.

Barry D, Hartigan JA. 1993. A Bayesian analysis for change point problems. *J. Am. Stat. Assoc.* **88**: 309–319.

Begert M, Schlegel T, Kirchhofer W. 2005. Homogeneous temperature and precipitation series of Switzerland from 1864 to 2000. *Int. J. Climatol.* **25**: 65–80.

Buishand TA. 1982. Some methods for testing the homogeneity of rainfall records. *J. Hydrol.* **58**: 11–27.

Caussinus H, Mestre O. 2004. Detection and correction of artificial shifts in climate series. *J. R. Stat. Soc. Series B* **C53**: 405–425.

Chow GC. 1960. Tests of equality between sets of coefficients in two linear regressions. *Econometrica* **28**(3): 591–605.

Corder GW, Foreman DI. 2009. *Nonparametric Statistics for Non-Statisticians*. John Wiley & Sons: Hoboken, NJ.

Costa AC, Soares A. 2009. Homogenization of climate data: review and new perspectives using geostatistics. *Math. Geosci.* **41**: 291–305.

DeGaetano AT. 2006. Attributes of several methods for detecting discontinuities in mean temperature series. *J. Clim.* **19**: 838–853.

Domonkos P. 2008a. Quantifying efficiency of homogenisation methods. In *Proceedings of the Sixth Seminar for Homogenization and Quality Control in Climatological Databases*, *Climate Data and Monitoring*, WMO-TD No. 1576, 20–32, World Meteorological Organization, Geneva, Switzerland.

Domonkos P. 2008b. Testing of homogenization methods: purposes, tools and problems of implementation. In *Proceedings of the 5th Seminar for Homogenization and Quality Control in Climatological Databases*, WCDMP-No. 71, WMO-TD 1493, WMO, Geneva, Switzerland, 126–145.

Domonkos P. 2011. Efficiency evaluation for detecting inhomogeneities by objective homogenization methods. *Theor. Appl. Climatol.* **105**: 455–467.

Domonkos P. 2013. Measuring performances of homogenization methods. *Q. J. Hungarian Meteorol. Serv.* **117**(1): 91–112.

Ducre-Robitaille JF, Vincent LA, Boulet G. 2003. Comparison of techniques for detection of discontinuities in temperature series. *Int. J. Climatol.* **23**: 1087–1101.

Easterling DR, Peterson TC. 1995. A new method for detecting undocumented discontinuities in climatological time series. *Int. J. Climatol.* **15**: 369–377.

Erdman C, Emerson JW. 2007. bcp: An R package for performing a Bayesian analysis of change point problems. *J. Stat. Softw.* **23**(3): 1–13.

Firat M, Dikbas F, Koc AC, Gungor M. 2010. Missing data analysis and homogeneity test for Turkish precipitation series. *Sadhana* **35**: 707–720.

Friedman M. 1937. The use of ranks to avoid the assumption of normality implicit in the analysis of variance. *J. Am. Stat. Assoc.* **32**(200): 675–701.

Gonzales-Rouco JF, Jimenez JL, Quesada V, Valero F. 2001. Quality control and homogeneity of precipitation data in the Southwest of Europe. *Am. Meteorol. Soc.* **14**: 964–978.

Guijarro JA. 2013. Climatological series shift test comparison on running windows. *Q. J. Hungarian Meteorol. Serv.* **117**(1): 35–45.

Hanssen-Bauer I, Førland E. 1994. Homogenizing long Norwegian precipitation series. *J. Clim.* **7**: 1001–1013.

Hawkins PM. 1977. Testing a sequence of observations for a shift in random location. *J. Am. Stat. Assoc.* **73**: 180–185.

Karabork MC, Kahya E, Komuscu AU. 2007. Analysis of Turkish precipitation data: homogeneity and the Southern Oscillation forcings on frequency distributions. *Hydrological Process* **21**: 3203–3210.

Karl TR, Williams CN. 1987. An approach to adjusting climatological time series for discontinuous inhomogeneties. *J. Clim. Appl. Meteorol.* **26**: 1744–1763.

Kruskal WH. 1952. A nonparametric test for the several sample problem. *Ann. Math. Stat.* **23**: 525–540.

Kruskal WH, Wallis WA. 1952. Use of ranks in one-criterion variance analysis. *J. Am. Stat. Assoc.* **47**: 583–621.

Kwiatkowski D, Phillips PCB, Schmidt P, Shin Y. 1992. Testing the null hypothesis of stationarity against the alternative of a unit root: how sure are we that economic time series have a unit root? *J. Econ.* **4**: 159–178.

Li S, Lund R. 2012. Multiple changepoint detection via genetic algorithms. *J. Clim.* **25**(2): 674–686.

Lindau R, Venema V. 2013. On the multiple breakpoint problem and the number of significant breaks in homogenization of climate records. *Q. J. Hungarian Meteorol. Serv.* **117**(1): 1–34.

Lund R, Reeves J. 2002. Detection of undocumented changepoints: a revision of the two-phase regression model. *J. Clim.* **15**: 2547–2554.

Lund R, Wang XL, Lu Q, Reeves J, Gallagher C, Feng Y. 2007. Changepoint detection in periodic and autocorrelated time series. *J. Clim.* **20**: 5178–5190.

Maronna R, Yohai VJ. 1978. A bivariate test for detection of systematic change in mean. *J. Am. Stat. Assoc.* **73**: 640–645.

Meteorological Organization World (WMO). 1996. *Guide to Meteorological Instruments and Methods of Observation, Part III – Chapter 3 Quality Management WMO-No. 8*. WMO: Geneva, Switzerland.

Mihajlovic D. 2006. Monitoring the 2003–2004 meteorological drought over Pannonian part of Croatia. *Int. J. Climatol.* **26**: 2213–2225.

Moberg A, Alexandersson H. 1997. Homogenization of Swedish temperature data, part II: homogenized gridded air temperature compared with a subset of global gridded air temperature since 1861. *Int. J. Climatol.* **17**: 35–54.

Ozcelik D. 1996. *Homogeneity Analysis of the Precipitation Series Located in Turkey*. MSc thesis, Istanbul Technical University, Istanbul, Turkey (in Turkish).

Peterson TC, Easterling DR. 1994. Creation of homogeneous composite climatological reference series. *Int. J. Climatol.* **14**: 671–679.

Peterson TC, Easterling DR, Karl TR, Groisman P, Nicholls N, Plummer N, Torok S, Auer I, Boehm R, Gullet D, Vincent L, Heino R, Tuomenvirta H, Mestre O, Szentimrey T, Salinger J, Førland EJ, Hanssen-Bauer I, Alexandersson H, Jones P, Parker D. 1998. Homogeneity adjustments of in situ atmospheric climate data: a review. *Int. J. Climatol.* **18**: 1493–1517.

Pettitt AN. 1979. A non-parametric approach to the change-point problem. *J. Appl. Stat.* **28**: 126–135.

Potter KW. 1981. Illustration of new test for detecting a shift in mean in precipitation series. *Mon. Weather Rev.* **109**: 2040–2045.

R Development Core Team. 2013. *R: A Language and Environment for Statistical Computing, Version 3.1.3*. R Foundation for Statistical Computing: Vienna, Austria. ISBN: 3-900051-07-0, http://www.R-project.org/ (accessed 9 March 2015).

Reeves J, Chen J, Wang XL, Lund R, Lu QQ. 2007. A review and comparison of changepoint detection techniques for climate data. *J. Appl. Meteorol. Climatol.* **46**: 900–915.

Sahin S, Cigizoglu HK. 2010. Homogeneity analysis of Turkish meteorological data set. *Hydrol. Processes* **24**: 981–992.

Said SE, Dickey DA. 1984. Testing for unit roots in autoregressive-moving average models of unknown order. *Biometrika* **71**: 599–607.

SAS Institute Inc. 2008. *SAS/ETS® 9.2 User's Guide*. SAS Institute Inc.: Cary, NC. http://support.sas.com/documentation/cdl/en/etsug/60372/PDF/default/etsug.pdf (accessed 1 March 2008).

Slonosky VC, Jones PD, Davies TD. 1999. Homogenization techniques for European monthly mean surface pressure series. *J. Clim.* **12**(8): 2658–2672.

Sneyers R, Tuomenvirta H, Heino R. 1998. Observational inhomogeneities and detection of climate change – the case of the Oulu (Finland) air temperature series. *Geophysica* **34**(3): 159–178.

Sprent P, Smeeton NC. 2007. *Applied Nonparametric Statistical Methods*, 4th edn. Chapman and Hall/CRC: Boca Raton, FL.

Staudt M, Esteban-Parra MJ, Castri-Diez Y. 2007. Homogenization of long-term monthly Spanish temperature data. *Int. J. Climatol.* **27**: 1809–1823.

Sugahara S, Porfirio da Rocha R, Ynoue RY, Bomfim da Silveira R. 2012. Homogeneity assessment of a station climate series (1933–2005) in the Metropolitan area of São Paulo: instruments change and urbanization effects. *Theor. Appl. Climatol.* **107**: 361–374.

Syrakova M, Stefanova M. 2009. Homogenization of Bulgarian temperature series. *Int. J. Climatol.* **29**: 1835–1849.

© 2015 Royal Meteorological Society

10970088, 2016, 1, Downloaded from https://rmets.onlinelibrary.wiley.com/doi/10.1002/joc.4329 by University Of Texas At, Wiley Online Library on [16/10/2024]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Szentimrey T. 1999. Multiple analysis of series for homogenization (MASH). In *Second Seminar for Homogenization of Surface Climatological Data*. WCDMP 41, WMO-TD 962, Szalai S, Szentimrey T, Szinell C (eds). WMO: Geneva, Switzerland, 27–46.

Tayanc M, Dalfes N, Karaca M, Yenigun O. 1998. A comparative assessment of different methods for detecting inhomogeneties in Turkish temperature data set. *Int. J. Climatol.* **18**: 561–578.

Toreti A, Kuglitsch FG, Xoplaki E, Della-Marta PM, Aguilar E, Prohom M, Luterbacher J. 2011. A note on the use of the standard normal homogeneity test (SNHT) to detect inhomogeneities in climatic time series. *Int. J. Climatol.* **31**: 630–632.

Toreti A, Kuglitsch FG, Xoplaki E, Luterbacher J. 2012. A novel approach for the detection of inhomogeneities affecting climate time series. *J. Appl. Meteorol. Climatol.* **51**: 317–326.

Tuomenvirta H. 2001. Homogeneity adjustments of temperature and precipitation series – Finnish and Nordic data. *Int. J. Climatol.* **21**: 495–506.

Tuomenvirta H. 2002. Homogeneity testing and adjustment of climatic time series in Finland. *Geophysica* **38**: 15–41.

Tuomenvirta H, Drebs A. 1994. Homogeneity testing and management of metadata in Finland. In *Climate Variations in Europe, Proceedings of the European Workshop*, Kirkkonummi, Finland, Heino R (ed). Publications of the Academy of Finland 3/94, Academy of Finland, Helsinki, 321–329.

Tuomenvirta H, Heino R. 1996. Climatic changes in Finland – recent findings. *Geophysica* **32**(12): 61–75.

Turkes M. 1999. Vulnerability of Turkey to desertification with respect to precipitation and aridity conditions. *Turk. J. Eng. Environ. Sci.* **23**: 363–380.

Turkes M, Koc T, Sarı F. 2008. Spatiotemporal variability of precipitation total series over Turkey. *Int. J. Climatol.* **29**: 1056–1074.

Venema V, Mestre O, Aguilar E, Auer I, Guijarro JA, Domonkos P, Vertacnik G, Szentimrey T, Stepanek P, Zahradnicek P, Viarre J, Müller-Westermeier G, Lakatos M, Williams CN, Menne M, Lindau R, Rasol D, Rustemeier E, Kolokythas K, Marinova T, Andresen L, Acquaotta F, Fratianni S, Cheval S, Klancar M, Brunetti M, Gruber C, Duran MP, Likso T, Esteban P, Brandsma T. 2012. Benchmarking monthly homogenization algorithms. *Clim. Past* **8**: 89–115.

Vincent LA. 1998. A technique for the identification of inhomegeneities in Canadian temperature series. *J. Clim.* **11**: 1094–1104.

Von Neumann J. 1941. Distribution of the ratio of the mean square successive difference to the variance. *Ann. Math. Stat.* **12**: 367–395.

Wang XL. 2008a. Accounting for autocorrelation in detecting mean shifts in climate data series using the penalized maximal t or F test. *J. Appl. Meteorol. Climatol.* **47**: 2423–2444.

Wang XL. 2008b. Penalized maximal F test for detecting undocumented mean shift without trend change. *J. Atmos. Oceanic Technol.* **25**: 368–384.

Wang XL, Wen QH, Wu Y. 2007. Penalized maximal t test for detecting undocumented mean change in climate data series. *J. Appl. Meteorol. Climatol.* **46**: 916–931.

Wijngaard JB, Kleintank AMG, Können GP. 2003. Homogeneity of 20th century European daily temperature and precipitation series. *Int. J. Climatol.* **23**: 679–692.

Yazici C, Purutcuoglu V, Yozgatligil C, Batmaz I, Bayramoglu K, Iyigun C. 2011. Homogeneity Analysis for Dependent Climate Data. In *7th International Statistical Conference (ISTKON)*, Antalya, Turkey, April 28–May 1, 2011.

Yazici C, Yozgatligil C, Batmaz I. 2012. A simulation study on the performances of homogeneity tests applied in meteorological studies. In *ICACM: International Conference on Applied and Computational Mathematics Ankara*, Turkey, Book of Abstracts, 93.

Young KC. 1993. Detecting and removing inhomogeneities from long-term monthly sea level pressure time series. *J. Clim.* **6**: 1205–1220.

Yozgatligil C, Purutcuoglu V, Yazici C, Batmaz I. 2010. Plausibility of the SNHT on Turkish precipitation data. In *7th National Symposium on Statistical Days*, *32–34*, 28–30 June 2010, Ankara, Turkey.

Yozgatligil C, Purutcuoglu V, Yazici C, Batmaz I. 2011. Validity of homogeneity tests for meteorological time series data: a simulation study, In *Proceedings of the 58th World Statistics Congress (ISI2011)*, Dublin, Ireland. August, 2011. http://2011.isiproceedings.org/papers/950636.pdf (accessed 27 March 2015).

Zeileis A, Kleiber C, Kramer W, Hornik K. 2003. Testing and dating of structural changes in practice. *Comput. Stat. Data. Anal.* **44**: 109–123.

© 2015 Royal Meteorological Society

10970088, 2016, 1, Downloaded from https://rmets.onlinelibrary.wiley.com/doi/10.1002/joc.4329 by University Of Texas At, Wiley Online Library on [16/10/2024]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License